

GloVe vs Word2Vec vs FastText

Extending Word Embedding Analysis Using the Brown Corpus

Objective

Continue the analysis from the previous assignment by introducing a third embedding method: GloVe (Global Vectors). Students will train their own GloVe embeddings on the Brown corpus and compare GloVe's performance with the previously trained Word2Vec and FastText models.

Part 1: Training a GloVe Model

Required Library

Use the Python package: `glove_python_binary` (or `glove_python`).

Task 1.1 — Data Preparation

Students must reuse the same dataset as in the previous assignment:

- All Brown corpus sentences
- The same manually added test sentences (Groups 1–6)

This ensures a fair comparison with Word2Vec and FastText.

Task 1.2 — Train a GloVe Embedding

Students should train a GloVe model with parameters comparable to previous models:

- `vector_size`: **100**
- `window`: **5**
- `learning_rate`: **0.05** (or default)
- `epochs`: **10**
- `min_count`: **1**

Task 1.3 — Save the Model

Students must save the trained GloVe embeddings to disk for later evaluation.

Part 2: Experiments and Evaluation

Students will evaluate GloVe on the same experiments used for Word2Vec and FastText.

Experiment 1: OOV (Out-of-Vocabulary) Behavior

Test OOV words:

teachable, unteacher, supercomputer, miscomputation, unnaturally

Tasks:

- Attempt to obtain vectors for each OOV word.
- Record whether GloVe can produce vectors (generally it cannot).
- Compare GloVe with Word2Vec and FastText.

Experiment 2: Rare Words

Rare words: unteachable, recomputation, unnaturalness

Common related words: teacher, computation, natural

Tasks:

- Compute similarity scores for:
(unteachable, teacher), (recomputation, computation), (unnaturalness, natural)
- Compare results between Word2Vec, FastText, and GloVe.

Experiment 3: Morphological Relationships

Morphological families:

Family 1: teach → teacher → teaching → teaches

Family 2: compute → computer → computation → computed → computational

Tasks:

- For 'teaching', list the top 5 most similar words using GloVe.
- Repeat for 'computation'.
- Compare with Word2Vec and FastText.

Experiment 4: Morphological Analogies

Analogy tasks:

teacher : teaching = computer : ?

natural : naturally = quick : ?

Tasks:

- Use GloVe's analogy method (or vector arithmetic).
- Compare results with Word2Vec and FastText.